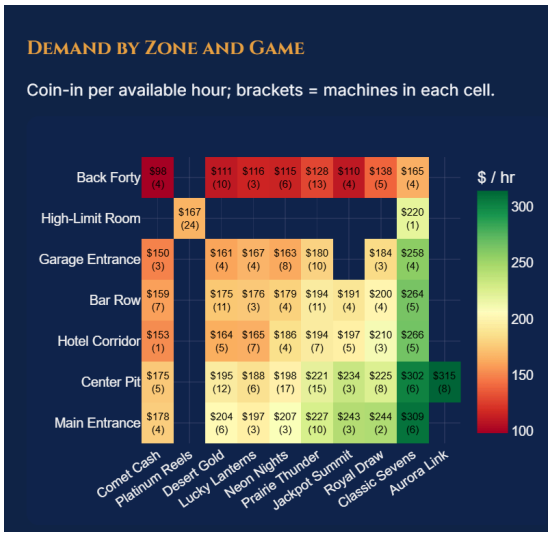
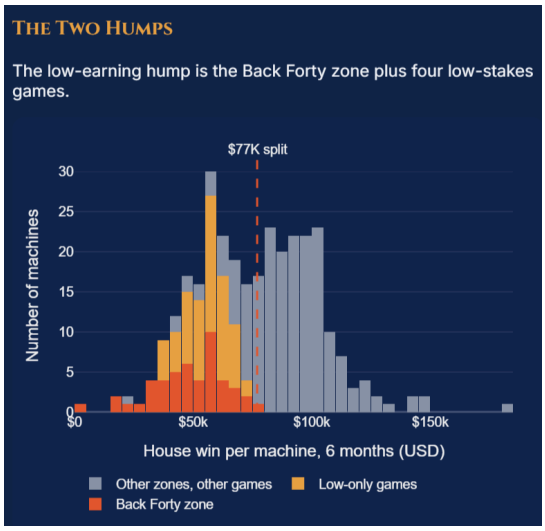
SUMMARY OF RESULTS

The manager wanted to replace the lowest earning machines. We believe that's wasted money. None of them is a "weak" machine: seven of them sit in the low traffic zone (Back Forty) and three were on the floor for only part of the 6 months. Location and game explain about 96% of the differences between machines. The machine itself explains 1%.





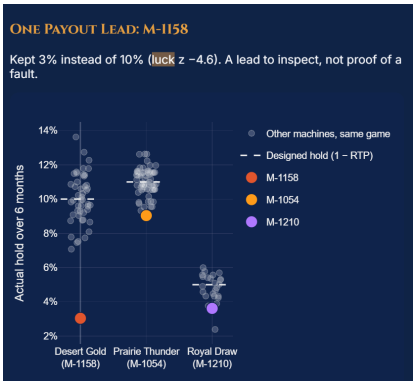
Luck, location or downtime?

We tested all three factors on all machines.

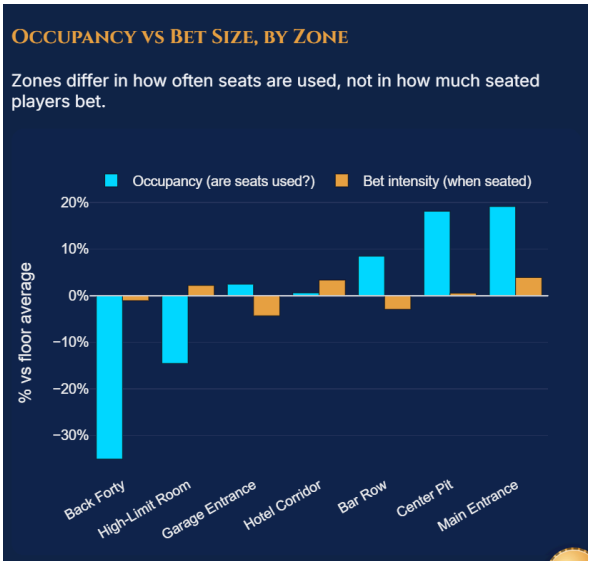
The floor kept 9.5% of coin-in against a designed 9.46% (weighted average). A luck z-score at the floor level did not help. At the machine level, however, M-1158 was flagged with a z-score of -4.6, keeping 3% instead of its 10% keeping design.

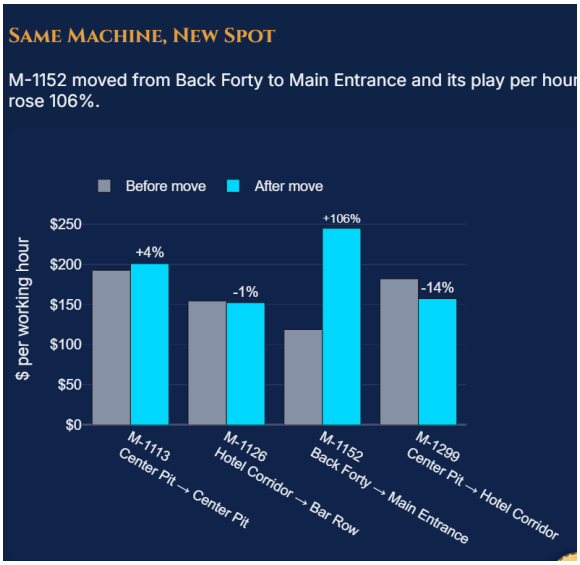
Location: Back Forty seats are mostly empty 35% more than the average of the floor. It is not a bet intensity problem, but a location one. M-1152 doubled its play (+106%) after it was moved to the main entrance.

Downtime: at the floor level, this cost 90K of expected win, under 0.4% of the total.



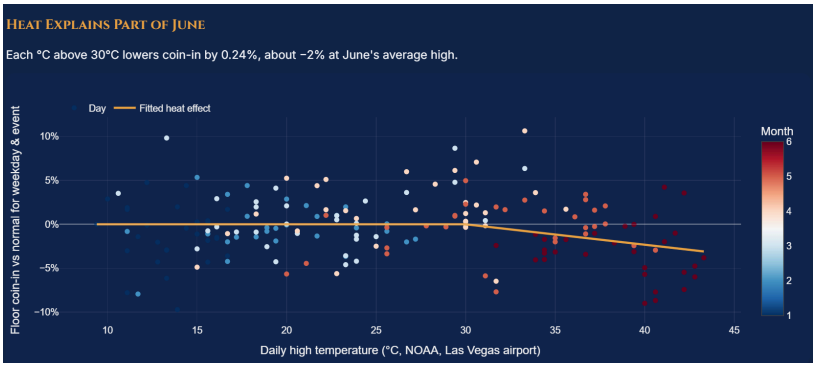
Demand, players and comps:
Demand was overall steady, with a weekly rhythm. Weekends which had sport events drove the spikes.
The top 10% of losing card players account for about 53% of all losses.
On May 1, comps became about 53% cheaper in points, comp cost rose about 8% but visits and play stayed flat.

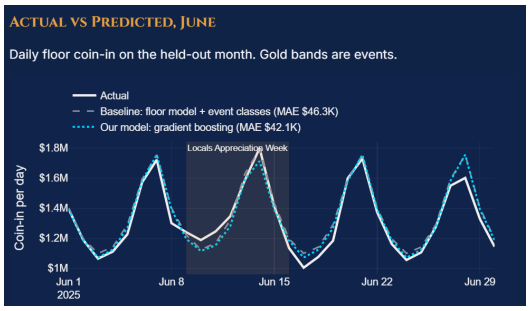


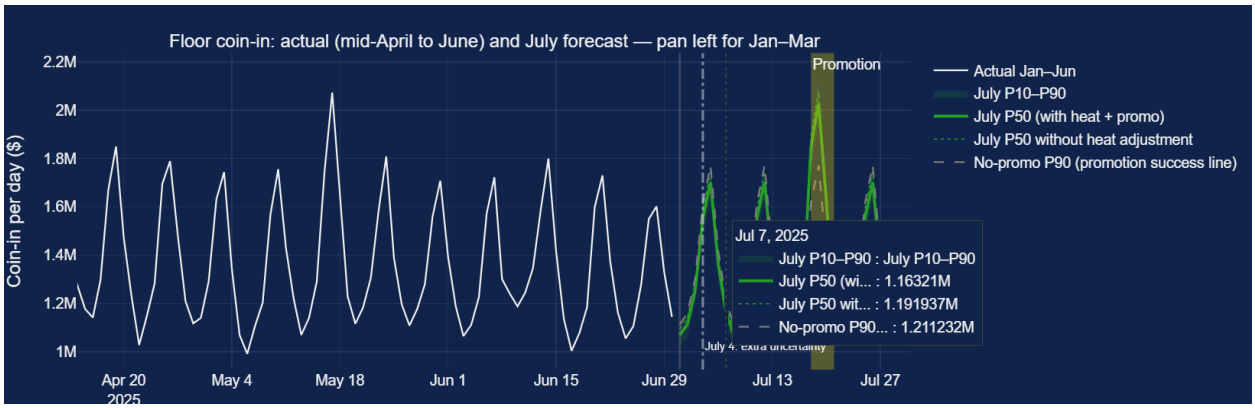


Forecast

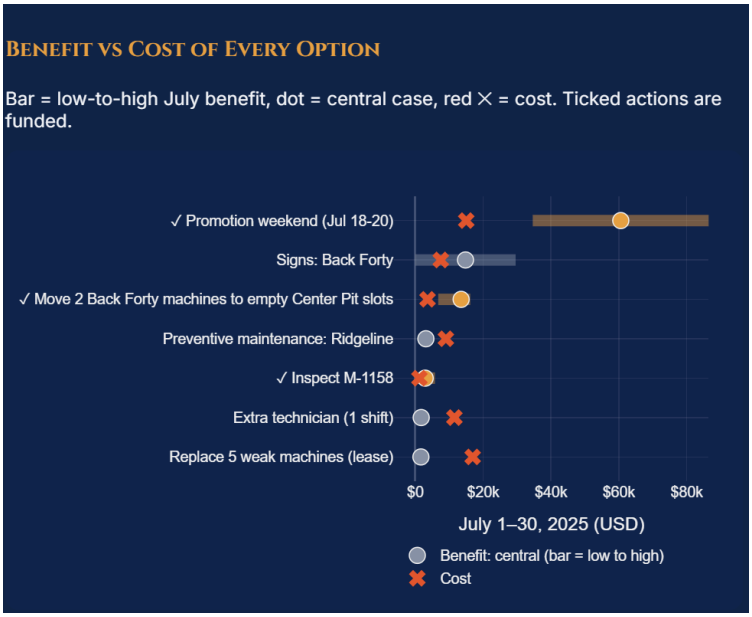
We tried several ML models, trained from January to May and tested in June. The best, gradient boosting on bank, game, weekday and event type, missed daily floor coin-in by $42.1K on average (MAE), against $46.3K for a simple weekday × event baseline: 9% better. Coin-in also dips on days above 30°C. That effect only shows once June's heat is in the data, so we apply it to July as an adjustment. Refit on all six months with the promotion added, the model forecasts July at $39.9M.







Recommendations:
Spend $19,800 and keep $80,200.



Action	Cost	July Benefit (conservative - central - optimistic)	How to measure?
Inspect M-1158	$1200	$0- 2.9K - 5.8K	July hold vs designed 10%
Move 2 Back Forty machines to Center Pit	$3600	$6.7 - $13.5 - $16.2	Coin-in per hour vs before move
Promo weekend (Jul 18-20)	$15000	$34.6 - $60.6K - $86.5K	Coin-in above $4.81 M (where the model assumes no promo (P90)

AI-use disclosure

We used **Claude (Anthropic)**, a generative AI assistant, during the datathon for the following:

- **Visualization code:** converting our matplotlib charts to interactive Plotly charts, styling them consistently, and adding zoom sliders and hover details.
- **Dashboard build:** writing and styling the HTML/CSS dashboard (layout, Compact/Detailed view, machine table, navigation) and embedding our chart files.